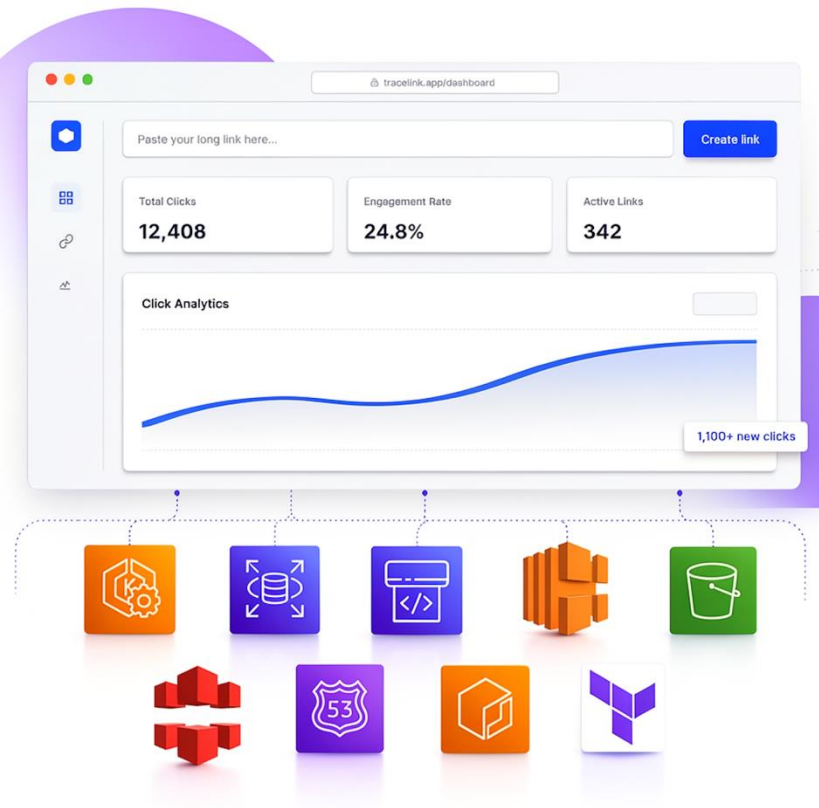




TraceLink

Enterprise-Grade URL Shortening & Analytics Platform

Cloud-Native · Scalable · Observable · Secure

A scalable cloud-native engineering project demonstrating production-grade architecture across AWS, Kubernetes, and modern CI/CD automation.

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SPRING BOOT

AWS

KUBERNETES

CI/CD

TERRAFORM

Agenda

A structured walkthrough of TraceLink's architecture, engineering decisions, and AWS infrastructure — from product concept to production deployment.

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Product Overview

What TraceLink is and the problem it solves

02

Core Features

URL shortening, analytics, QR codes, APIs & email

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Architecture Overview

End-to-end AWS cloud-native system design

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AWS Services Deep Dive

ECR, RDS, EKS, Route53 & CloudFront

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Zero-downtime deployment automation

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Results & Learnings

Key outcomes, engineering takeaways, and next steps

What Is TraceLink?

A Production-Grade URL Management Platform

TraceLink is a full-stack, cloud-native URL shortening and analytics platform built for enterprise-scale workloads. It combines a modern React frontend with a Spring Boot REST API backend, deployed on Amazon EKS with managed infrastructure provisioned entirely via Terraform. Every component is containerised, observable, and designed for zero-downtime operation.

URL Shortening

Custom aliases, expiration dates, and high-throughput redirection

Real-Time Analytics

Click tracking, device detection, and geographic insights

QR Code Generation

Dynamic QR codes generated per short URL on demand

REST APIs

Documented, versioned APIs with Bearer token authentication

Email Workflows

Gmail SMTP integration for notifications and verification

BACKEND: JAVA 21 + SPRING BOOT

FRONTEND: REACT 19 + VITE

DATABASE: MYSQL 8

CLOUD: AWS EKS + RDS + S3 + CLOUDFRONT

Core Features

TraceLink delivers a comprehensive feature set designed for both end-user experience and developer integration — all backed by production-grade security and observability.

Create new link

Paste your long URL to shorten and track it.

Destination URL

+ Add UTM parameters

CancelShorten link

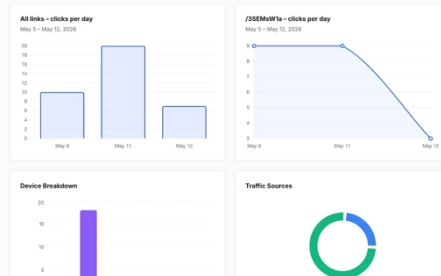
URL Shortening

Generate short URLs with custom aliases and configurable expiration dates. Supports bulk operations and link lifecycle management.



QR Code Generation

Automatically generate downloadable QR codes for every short URL, enabling seamless offline-to-digital campaigns.



Click Analytics

Real-time dashboards tracking clicks, referrers, device types, and geographic distribution of traffic.

Developers

Integrate TraceLink directly into your application using our REST API.

API Keys

Manage your secret API keys. Use these keys to authenticate your API requests. Keep them secure and never expose them in client-side code.

Your new API key is ready

Important: This key will only be shown once. Store it securely in your environment.

`tl_live_05139hjE22gnRbeeJ5xgc1A542NvHfyR0PqFzI`

I have saved this key

Standard Keys

NAME	KEY	CREATED
ServerKey	<code>tl_live_***+0FrI</code>	May 11, 2020

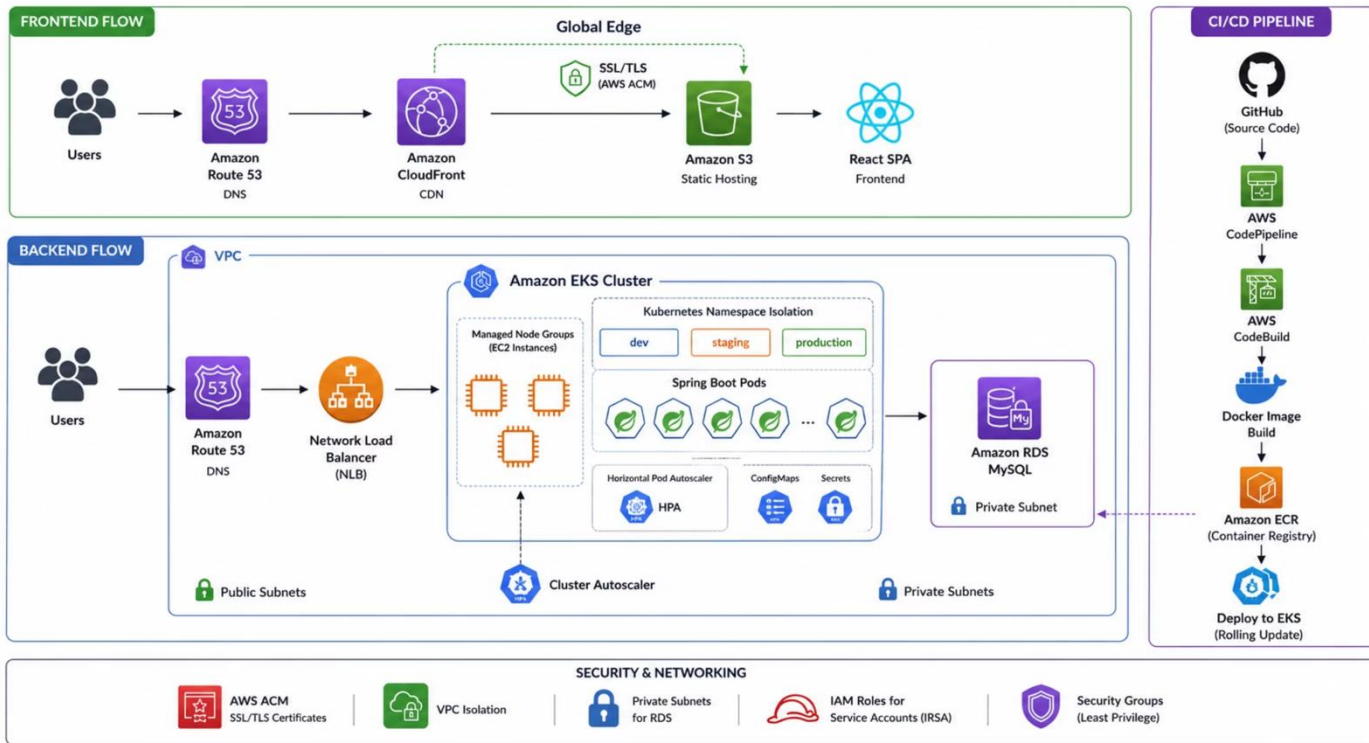
API Keys & Auth

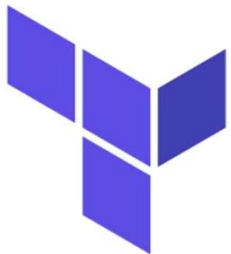
BCrypt-hashed API keys with Bearer token authentication. Secure, scoped access for programmatic integrations.

Complete Cloud Architecture

End-to-End AWS Infrastructure Design

TraceLink's architecture is fully cloud-native, leveraging managed AWS services orchestrated through Kubernetes. The system is split into two distinct traffic paths – a static frontend served via CloudFront and S3, and a dynamic backend running on Amazon EKS behind a Network Load Balancer. All infrastructure is provisioned via Terraform, ensuring repeatable, version-controlled deployments.





Infrastructure Provisioning with Terraform

Automated, Reproducible & Scalable AWS Infrastructure

TraceLink's entire AWS infrastructure is defined and managed using Terraform, ensuring every environment — from development to production — is identical and provisioned automatically. This approach eliminates manual configuration errors and significantly accelerates deployment cycles.

Declarative Configuration

Infrastructure defined as code (HCL), describing the desired state of AWS resources.

Reproducible Environments

Ensures consistent AWS infrastructure across all stages (dev, staging, production).

Version Control

Infrastructure changes are tracked, reviewed, and approved just like application code.

Automated Deployments

Terraform applies changes, preventing drift and enabling rapid scaling.

Terraform Workflow



```
# RDS Instance
resource "aws_db_instance" "tracelink-db" {
  identifier = "tracelink-db"

  # Engine
  engine      = "mysql"
  engine_version = "8.0"

  # Compute & Storage
  instance_class      = var.db_instance_class
  allocated_storage   = var.db_allocated_storage
  storage_type         = "gp3"
  storage_encrypted    = true # Always encrypt at rest in production

  # Database
  db_name      = var.db_name
  username     = var.db_username
  password     = var.db_password

  # High Availability
  multi_az = var.db_multi_az

  # Networking
  db_subnet_group_name = aws_db_subnet_group.tracelink.name
  vpc_security_group_ids = [aws_security_group.rds_sg.id]
  publicly_accessible    = true # Private - only accessible from within VPC

  # Maintenance & Backups
  backup_retention_period = var.db_backup_retention_days
  apply_immediately       = true # Apply changes immediately for demo

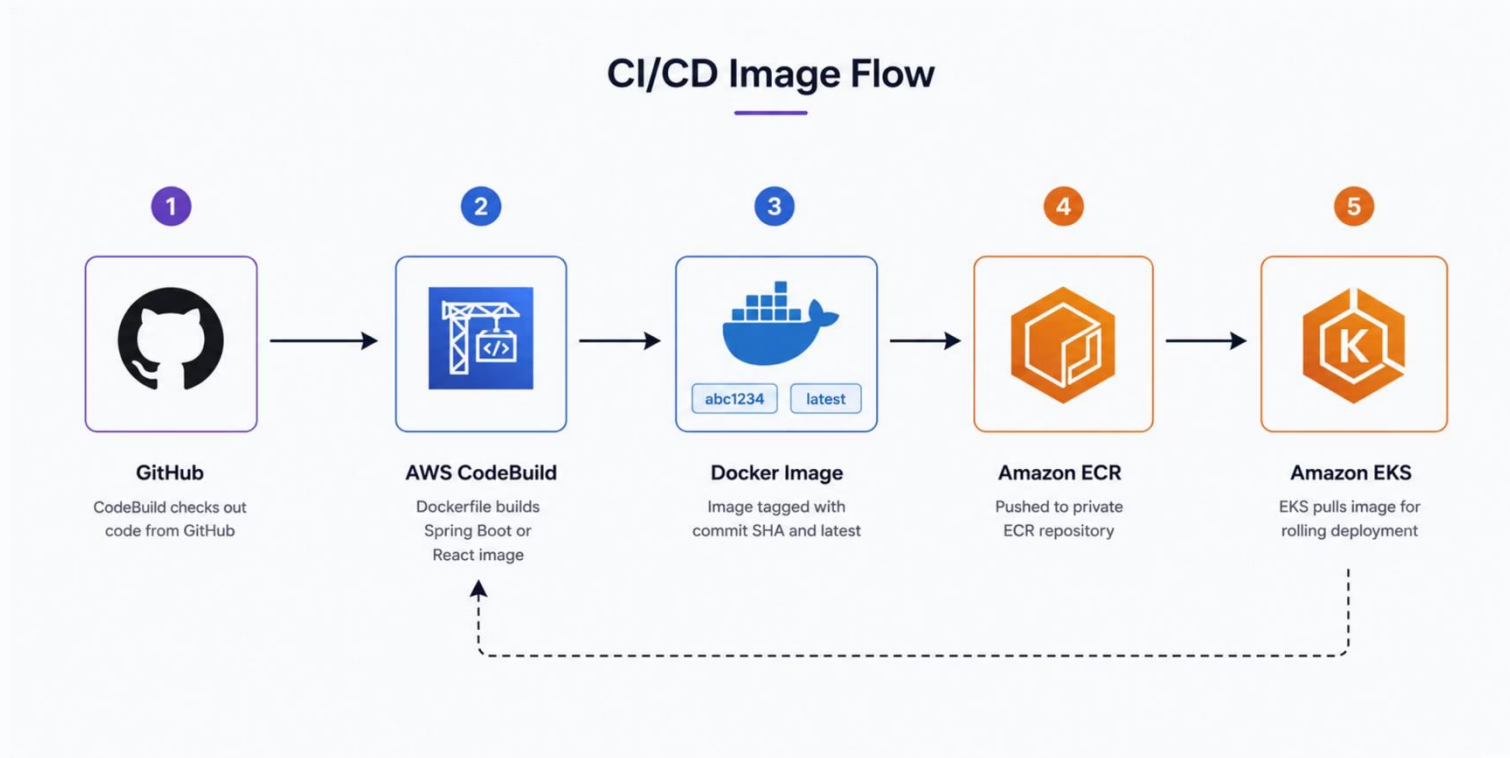
  # Lifecycle
  skip_final_snapshot = true # In prod, always take a final snapshot before

  tags = {
    Name = "tracelink-db"
  }
}
```

Amazon Elastic Container Registry

Secure, Versioned Docker Image Storage

Amazon ECR serves as the private container registry for all TraceLink workloads. Every Docker image is built during the CI/CD pipeline, scanned for vulnerabilities using AWS-native tools, tagged with Git commit SHAs for traceability, and pushed to ECR before deployment to EKS. Lifecycle policies automatically expire stale images, keeping the registry clean and cost-efficient.



Amazon RDS MySQL 8



Managed, Encrypted, Highly Available Database

Amazon RDS provides a fully managed MySQL 8 database for TraceLink's persistence layer. It runs in a private VPC subnet, is only reachable from EKS pods over JDBC, and includes automated backups, point-in-time recovery, and Multi-AZ replication for durability and availability.

Security & Isolation

Private subnet, no public endpoint, SSL JDBC, and credentials in Kubernetes Secrets.

High Availability

Multi-AZ failover, daily snapshots, and point-in-time recovery.

Performance

MySQL 8 tuned for read-heavy workloads with HikariCP pooling.

Infrastructure as Code

Provisioned with Terraform for repeatable environments.

Kubernetes Orchestration with Amazon EKS



Managed Kubernetes for Production Workloads

Amazon EKS runs TraceLink's Spring Boot backend as containerised workloads. The cluster uses managed node groups with autoscaling enabled, Horizontal Pod Autoscaling for demand-based scaling, and rolling deployment strategies for zero-downtime releases. ConfigMaps and Kubernetes Secrets manage environment-specific configuration securely.

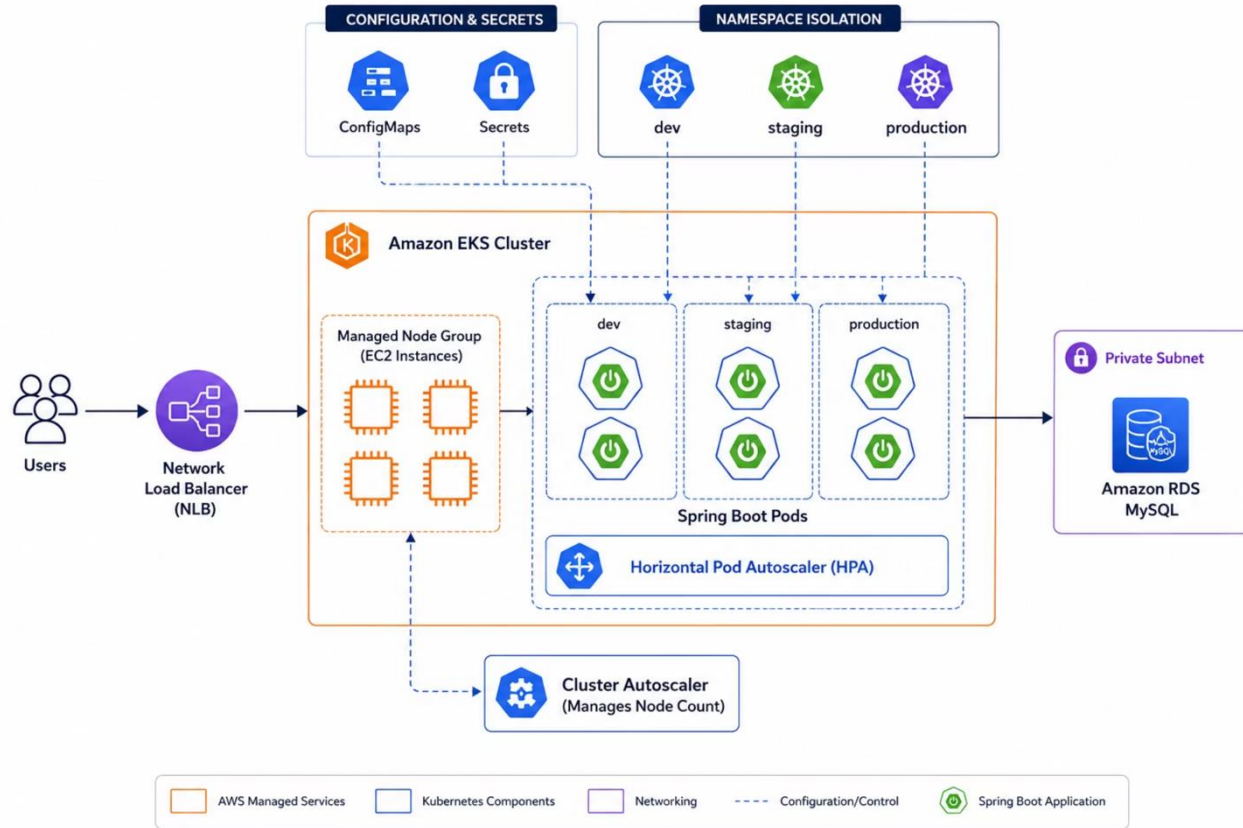
Cluster Architecture

- **Managed Control Plane:** AWS operates the EKS control plane
- **Node Groups:** EC2-based worker nodes with autoscaling
- **HPA:** Horizontal Pod Autoscaling based on CPU/memory metrics
- **Namespaces:** Isolated environments for dev, staging, and production
- **Secrets & ConfigMaps:** Secure, non-hardcoded configuration

Scaling & Resilience

- Rolling deployments with zero downtime
- Readiness and liveness probes for health checks
- Pod Disruption Budgets for controlled maintenance
- Cluster Autoscaler for node-level scaling
- Network Policies for pod-to-pod traffic control

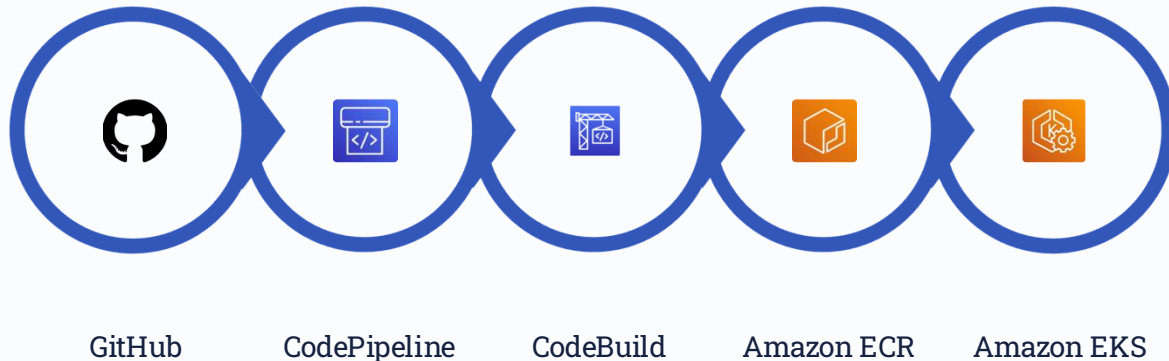
Amazon EKS & Kubernetes



Zero-Downtime CI/CD Pipeline

AWS CodePipeline + CodeBuild + EKS Rolling Deployments

TraceLink's CI/CD pipeline automates the entire release process from code commit to production deployment. Every push to the main branch triggers a validated, fully automated pipeline — building Docker images, scanning for vulnerabilities, pushing to ECR, and deploying to EKS with rolling updates. No manual intervention required.



The pipeline enforces quality gates at every stage — unit tests must pass in CodeBuild before the image is pushed, and Kubernetes readiness probes confirm pod health before traffic is shifted during rolling deployments.

Pipeline Features

- Automated on every main-branch commit
- Rolling deployment strategy with zero downtime
- Automated rollback on health-check failure
- Git SHA image tagging for full traceability

Tools & Integration

- **AWS CodePipeline** — Pipeline orchestration
- **AWS CodeBuild** — Build, test and deployment execution
- **Amazon ECR** — Container image registry
- **Amazon EKS** — Kubernetes deployment target

Results & Key Learnings

TraceLink demonstrates a fully operational, production-grade cloud-native platform — built, deployed, and automated entirely on AWS using Infrastructure as Code and Kubernetes best practices.

Infrastructure as Code

100% of AWS resources provisioned via Terraform — repeatable, version-controlled, and auditable environments.

Zero-Downtime Deployments

Rolling updates on EKS with health probes ensure continuous availability during every release cycle.

Secure Cloud Networking

Private VPC subnets, ACM SSL/TLS, encrypted RDS connections, and IAM least-privilege roles throughout.

Observability & Scaling

HPA, Cluster Autoscaler, CloudFront CDN, and structured logging provide production-grade observability.

7+

AWS Services

EKS, RDS, ECR, S3, CloudFront, Route53, CodePipeline, CodeBuild, ACM

0

Downtime Deployments

Rolling update strategy with Kubernetes readiness probes

100%

IaC Coverage

All infrastructure defined and managed via Terraform

Thank you for your attention. TraceLink represents a comprehensive demonstration of cloud-native engineering — combining full-stack development, AWS architecture, Kubernetes orchestration, and CI/CD automation into a single, cohesive production-grade platform.